Tender: Development of Centralized Access Control Integration System — B. BIRDS (Border Incident Reporting & Documentation System) - Phase II
e-GP tender tracking no.: 1347267
Date: 18/10/2026

To: Director (Communication), Communication Branch, Headquarters Border Guard Bangladesh, Peelkhana, Dhaka

TECHNICAL PROPOSAL

BIRDS (Border Incident Reporting & Documentation System) - Phase II

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1. Introduction

Grace International submits this technical proposal for the supply, installation, commissioning and support of the Border Incident Reporting & Documentation System (BIRDS) Phase II for Headquarters Border Guard Bangladesh under e-GP tender 1347267 (Development of Centralized Access Control Integration System — B. BIRDS Phase II). It describes the offered system, its architecture and security, the infrastructure it needs, how availability, backup and recovery are achieved, and how the system will be implemented, tested, handed over and supported.

The offered system is already built and running. The Technical Evaluation Committee can verify it in the application demo (address and accounts in Annex A, section 1). Compliance with each of the 58 items of the Technical Specification is stated in the Compliance Statement and explained in Annex A; features offered beyond the specification are listed in Annex B.

2. Company profile and project team

The tenderer is Grace International. Its registration, licences, experience records and other verification documents are submitted separately with this tender, together with the curricula vitae of the key personnel.

The project is staffed with the following roles throughout implementation and the warranty period:

Role	Responsibility
Project manager	Single point of contact for BGB; plan, progress reporting, risks, acceptance
Solution architect	Architecture, integration (directory, e-mail, offline map data, antivirus), performance and availability design
Software engineers	Configuration and customisation of forms, reports and workflow; fixes during warranty
Database / systems administrator	Servers, database replication, backup and restore, DR site, monitoring
Security engineer	Hardening, security testing and vulnerability assessment, remediation
Trainer / support lead	Training courses and material, user support and the support desk

3. Understanding of the requirement

BGB needs one central system in which incidents along the border are reported by the units that observe them, checked and approved through the chain of command, and then tracked, analysed and reported at every level from HQ to BOP. The record must be trustworthy: each step attributed, nothing finalised altered, evidence preserved with its original metadata, and every significant action auditable. The system must run on BGB's own network with high availability, recover from failure, integrate with BGB's directory, and remain open to future systems such as mobile applications, surveillance and digital evidence systems.

4. Solution overview

BIRDS is a web application used through a browser on BGB's network; no software has to be installed on users' computers. It is also a progressive web app (PWA), so it can be installed on phones, tablets and computers directly from the browser (section 5.3). Its modules are:

Module	What it does	Spec items
Incident registration	Three-step electronic form, drafts, submission, editing after return, duplicate warning, GPS location, evidence upload (photos, video, PDF, office documents)	5–11
Workflow and verification	Draft → Submitted → Reviewed → Verified → Approved → Closed, with Return and Reject; modification by the reviewer before approval (reason required, audited); role- and scope-based actions; overdue flags; timeline and comments	12–14, 25
Tracking and search	Unique reference numbers BIRDS-<year>-NNNN, advanced filters, saved views, global search	15, 16
Dashboard, GIS and analytics	Role-based dashboard and work queue, incident map with per-BOP counts, zoom to a location and drill-down to the incidents, clustering and heat map, statistics with hotspots and time-of-day patterns, trends by day/week/month/quarter/year, every chart opening its incidents	17–20
Reports and export	Configurable report templates, formal printouts with reference numbers, PDF, Excel and CSV export	21, 22
Notifications	In-app, e-mail and push notifications for workflow events	23
Audit and integrity	Tamper-evident audit trail, append-only history, evidence fingerprints	24, 25
Administration	Users, roles and permissions, organisational hierarchy, categories, workflow, security and retention settings	3, 6, 9, 46, 47
Archiving and retention	Configurable retention and archiving with audit trail	44, 45
Integration	Documented REST API; directory (AD/LDAP) sign-in	31, 32, 58

4.1 Incident workflow

Every incident follows the workflow below (items 13, 14). The server holds the only copy of these rules: it decides which actions each officer may take on each record, so the screens only offer what the server will accept. Each step notifies the next responsible officers (item 23), is added to the incident timeline (item 12) and is recorded in the audit trail (item 24).



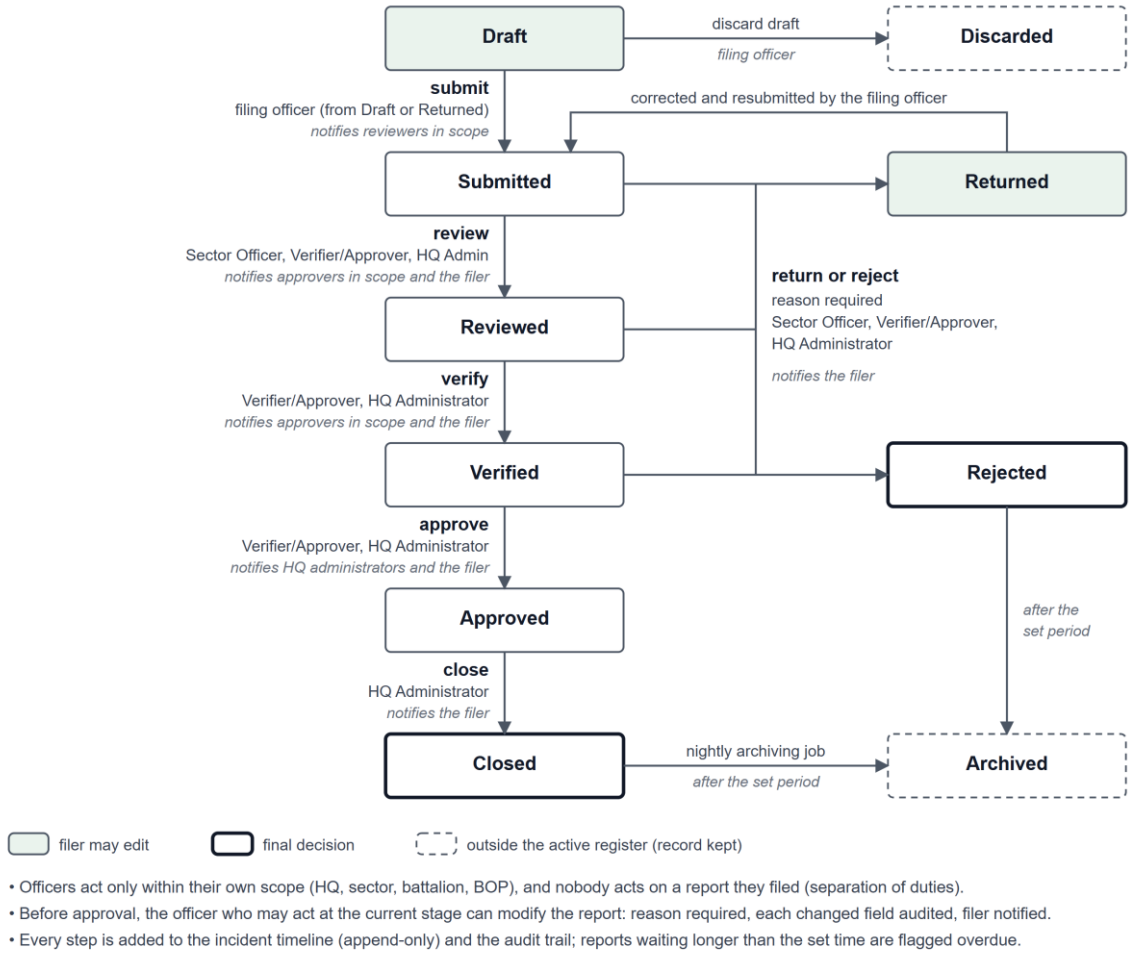


Figure 1. Incident workflow: states, who may act at each step, and who is notified (items 12, 13, 14, 23, 25)

4.2 Roles, permissions and scope

Access is decided by two things: the pages and actions the user's role is permitted (item 3), and the user's scope in the organisational hierarchy (item 9). The default permissions are shown below; administrators change them in Administration › Permissions without code changes (items 46, 47), and the server enforces them on every request.

Page / function	HQ Administrator	Sector Officer	Battalion Operator	Verifier / Approver	Auditor
Dashboard	Yes	Yes	Yes	Yes	Yes
Incident register and records	Yes	Yes	Yes	Yes	Yes
File new incidents	Yes	Yes	Yes	—	—
GIS map	Yes	Yes	Yes	Yes	—

Page / function	HQ Administrator	Sector Officer	Battalion Operator	Verifier / Approver	Auditor
Analytics	Yes	Yes	—	Yes	Yes
Reports and export	Yes	Yes	—	Yes	Yes
Audit trail	Yes	—	—	—	Yes
Administration	Yes	—	—	—	—
Messages and voice calls	Yes	Yes	Yes	Yes	Yes
Raise flash alerts	Yes	Yes	—	—	—
Workflow actions	Review, verify, approve, close, return, reject	Review, return, reject	Submit own reports	Review, verify, approve, return, reject	None (read only)

Each user is assigned a scope: HQ (every unit), a sector, a battalion or a BOP. Lists, the map, dashboards, analytics, reports, exports and workflow actions include only the incidents within that scope, and a user who is not permitted a record receives "not found" rather than its contents. The hierarchy HQ → Sector → Battalion → BOP is maintained in Administration › Organisation. Separately from scope, nobody can review, verify or approve a report they filed themselves.

4.3 Evidence handling

Evidence files pass through the steps below before they are stored, and are checked again every time they are opened (items 10, 11, 25, 33). The original metadata is kept with the record, so the evidential value of photographs and documents is preserved.



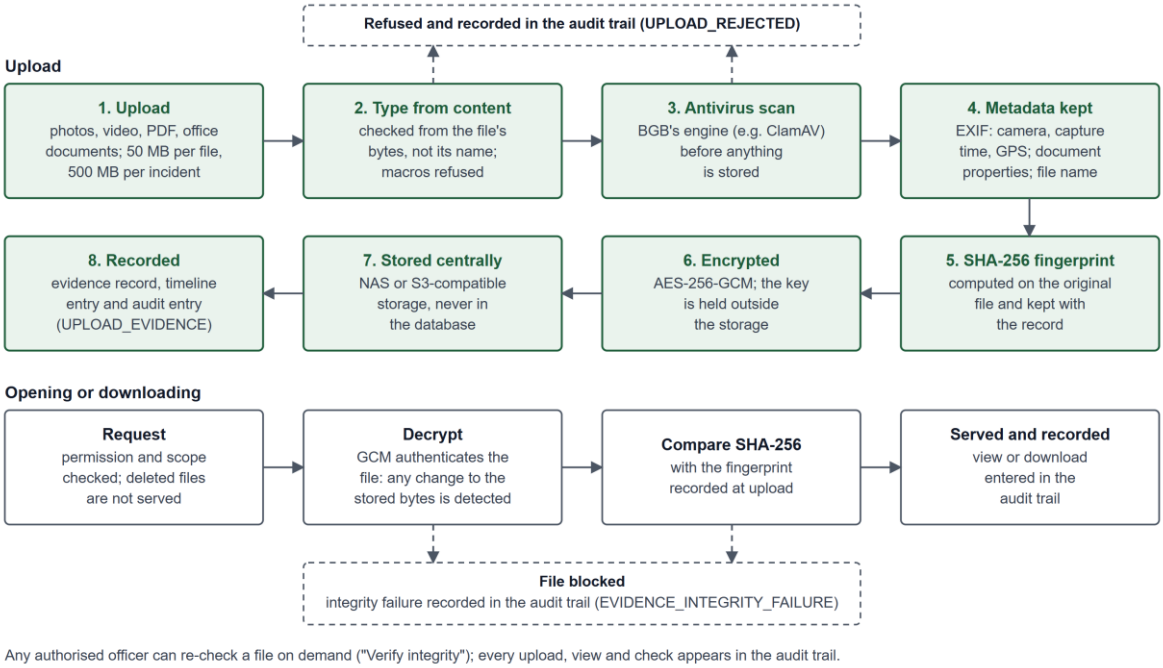


Figure 2. Evidence upload and retrieval with integrity checks (items 10, 11, 25, 33)

5. System architecture

5.1 Three-tier design

BIRDS follows the three-tier architecture required by item 2. Each tier can be scaled and secured separately.

Tier	Technology	Role
Presentation	React 19 single-page application built with Vite; Leaflet maps; Chart.js charts; English and Bangla	Runs in the user's browser (Chrome, Edge, Firefox); progressive web app installable from the browser on phones, tablets and computers
Application	Node.js (LTS, version 20 or later), Express 5, TypeScript; Zod validation; WebSocket for realtime events	All business rules, permissions, workflow, audit and file handling; REST API at /api/v1
Data	PostgreSQL 16 or later (relational, ACID); file storage on disk/NAS or S3-compatible object storage	Incident records, history, audit trail, configuration; encrypted evidence files

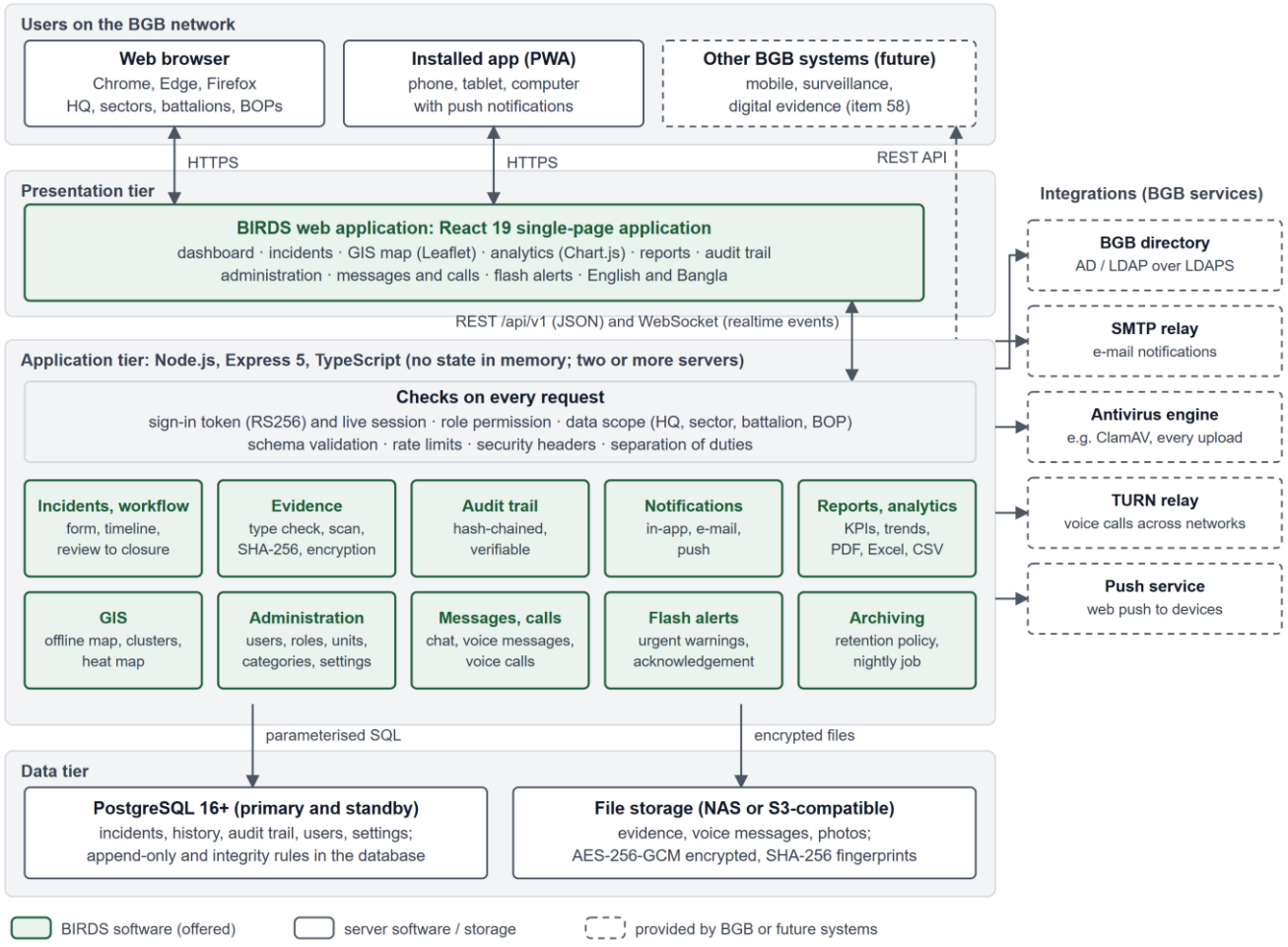


Figure 3. Solution architecture: three tiers, application modules and integrations (items 2, 30, 31, 58)

5.2 Deployment layout

Users connect over HTTPS to a reverse proxy (Nginx) that terminates TLS and balances requests across two or more application servers. The application servers hold no state in memory: sessions are stored in the database and realtime events are passed through the database, so any server can answer any request and servers can be added or removed without interruption. The database runs as a primary with a standby replica; files are kept on central storage. A disaster-recovery site holds a replica of the database and files.

Component	Purpose
Reverse proxy / load balancer (Nginx)	TLS termination, HTTP→HTTPS redirect, security headers, load balancing, WebSocket forwarding
Application servers (×2 or more)	BIRDS API and web application, run under a process manager (PM2) using all CPU cores; they also hold the offline map data (section 7)

Component	Purpose
Database (primary + standby)	PostgreSQL with streaming replication; point-in-time recovery through WAL archiving
File storage	Central NAS or S3-compatible object storage (e.g. MinIO) for evidence, voice messages and photos; encrypted by the application before storage
Backup storage	Separate storage for encrypted database and file backups
Supporting services	SMTP relay (e-mail notifications), TURN relay (voice calls), antivirus engine for uploads
DR site	Database and file replicas with application servers ready to start

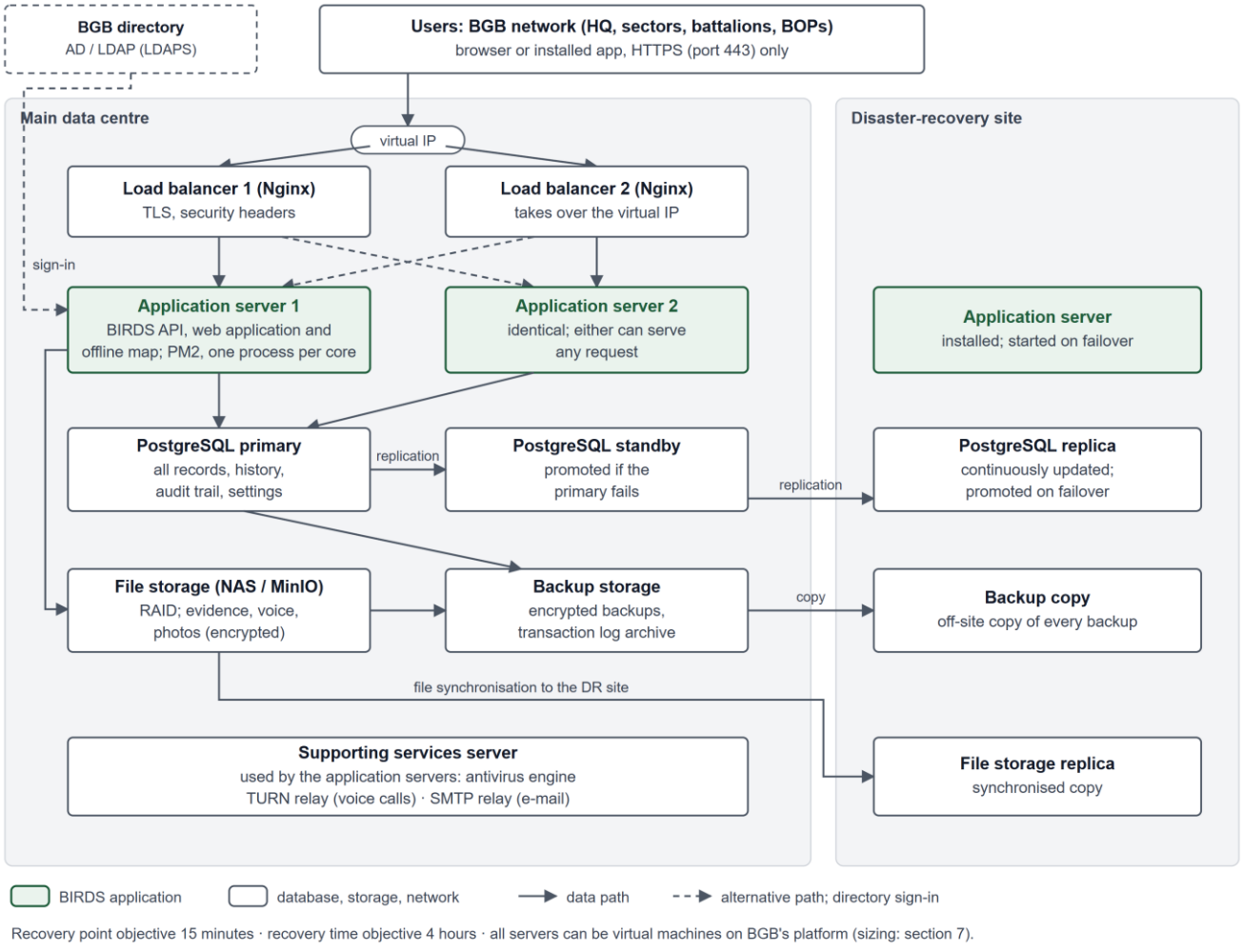


Figure 4. Deployment on BGB's network: main data centre and disaster-recovery site (items 27, 35, 36, 37); sizing in section 7

5.3 Mobile access: progressive web app

BIRDS is a progressive web app (PWA). The same application that runs in the browser can be installed on a phone, tablet or computer directly from the browser, without an app store and without a separate mobile application to develop, approve or maintain.

Aspect	How it works
Installation	Android, Windows, Mac: Chrome or Edge › "Install app" (also in the BIRDS user menu). iPhone/iPad: Safari › Share › "Add to Home Screen". BGB can also deploy it to managed devices.
Use	Opens from the home screen in its own full-screen window like a native app, with the BIRDS icon; the layout adapts to phone and tablet screens, including notched phones.
Updates	Every installed copy updates itself the next time it opens after the server is updated; there is nothing to reinstall.
Notifications	Calls and messages reach the device as push notifications even when BIRDS is closed (iPhone/iPad: iOS 16.4 or later, after installation).
Security	Only the application files are kept on the device; incident data is always fetched from the server after sign-in and is never stored on the device. Without a connection BIRDS opens with an offline notice.
Requirement	An HTTPS address with a certificate trusted by the devices, and network access from the devices to the BIRDS server (Annex B, section 3).

5.4 Software licensing

All components (Node.js, PostgreSQL, Nginx, PM2, React, Leaflet, MinIO or equivalent, coturn) are open source. No proprietary licence is needed to run BIRDS; BGB is not tied to any vendor's licence fees.

5.5 Database design

BIRDS uses PostgreSQL, an enterprise relational database (item 30), with 25 tables. The schema is managed as versioned migrations, so every installation has exactly the same structure and upgrades are repeatable. Figures 5 and 6 show the tables and their relationships; to keep them readable, the many columns that refer to a user are marked "→ users" instead of being drawn as lines. The full table and column dictionary is part of the Database Documentation (section 14).

a. Integrity rules in the database itself, so that they hold even outside the application: allowed values for severity, status and other fixed lists; unique reference numbers; foreign keys; triggers that refuse any change or deletion of the incident timeline, comments, messages and audit trail; a trigger that chains each audit entry to the previous one with SHA-256 (items 24, 25).

b. Indexes for every common search and dashboard: incident status, sector, date and category, sector with status, filer with status, archive date, and status with time in status (overdue); history, comments and evidence by incident; audit trail by time, user and action; notifications by user and read state; messages by conversation; calls by participant and time; flash alerts by status and validity. Every list is paged, so response time does not grow with the size of the register (items 37, 39).

c. Evidence and other files are kept in file storage, not in the database; the database holds their name, type, size, fingerprint and metadata. This keeps the database small and its backups fast (items 33, 34).



Figure 5. Entity-relationship diagram (1 of 2): organisation, users, incidents, evidence, audit trail (item 30)

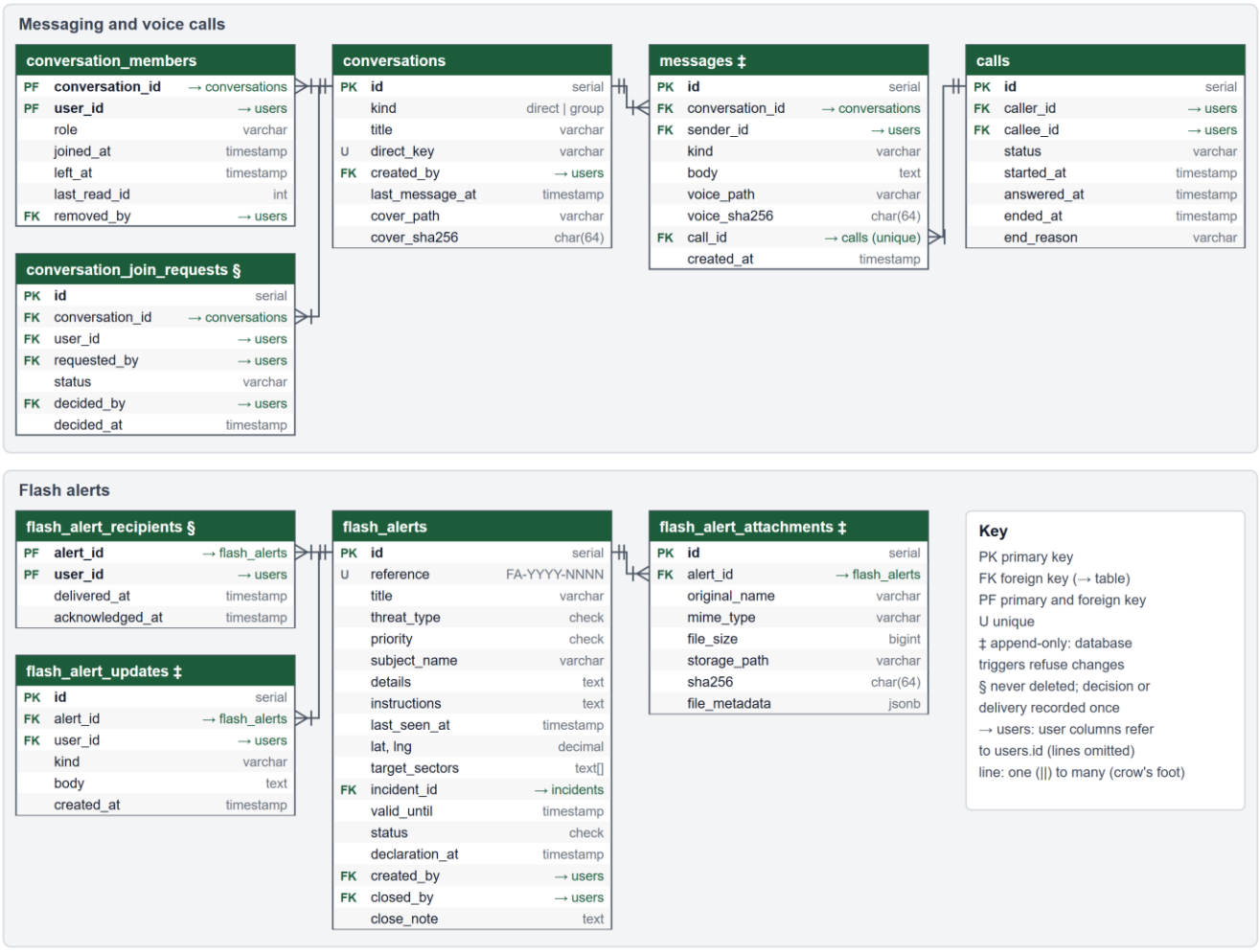


Figure 6. Entity-relationship diagram (2 of 2): messaging, voice calls and flash alerts (item 30)

6. Security architecture

Security is built into each layer, following recognised secure development practice and the OWASP Top 10 (item 49).

Area	Measures in the offered system
Authentication	Username and password; passwords hashed with bcrypt (cost 12); configurable length and complexity; lockout after a configurable number of failed attempts; forced change after an administrator reset; directory (AD/LDAP) sign-in at deployment

Area	Measures in the offered system
Sessions	Signed tokens (RS256) valid 15 minutes, renewed through an httpOnly cookie; the access token is held in memory only; configurable idle timeout with automatic sign-out; one active session per user; every request re-checks the session, so sign-out and deactivation take effect immediately
Authorisation	Role-based permissions per page and action, editable in the Admin module; data scope by HQ, Sector or Battalion; enforced by the server on every request, never only in the browser; separation of duties (nobody approves their own report)
Data protection	HTTPS/TLS for all traffic; evidence files, voice messages and photos encrypted at rest with AES-256-GCM; database storage encryption on the BGB servers
Integrity	The filer edits only in Draft/Returned; before approval a reviewer may modify a report with a recorded reason, each changed field audited with its old and new value; finalised records cannot be altered or deleted; history, comments and audit tables append-only by database triggers; audit trail chained with SHA-256 and verifiable on demand; evidence fingerprinted (SHA-256) and re-checkable
Input and uploads	Every request validated against a schema; database access only through parameterised queries; file type decided from the file's content, not its name (photos, video, PDF and office documents; macro-enabled documents refused); 50 MB per file and 500 MB per incident (configurable); antivirus scan hook for BGB's chosen engine (e.g. ClamAV)
Hardening	Security headers (Content Security Policy, frame, content-type, referrer), CORS limited to BGB addresses, rate limits (10 failed sign-ins per 15 minutes per address; 100 requests per minute per user), generic sign-in errors, dependency vulnerability audit
Monitoring	Separate application, HTTP and security logs; failed sign-ins, lockouts and permission denials recorded as security events; full audit trail in the application

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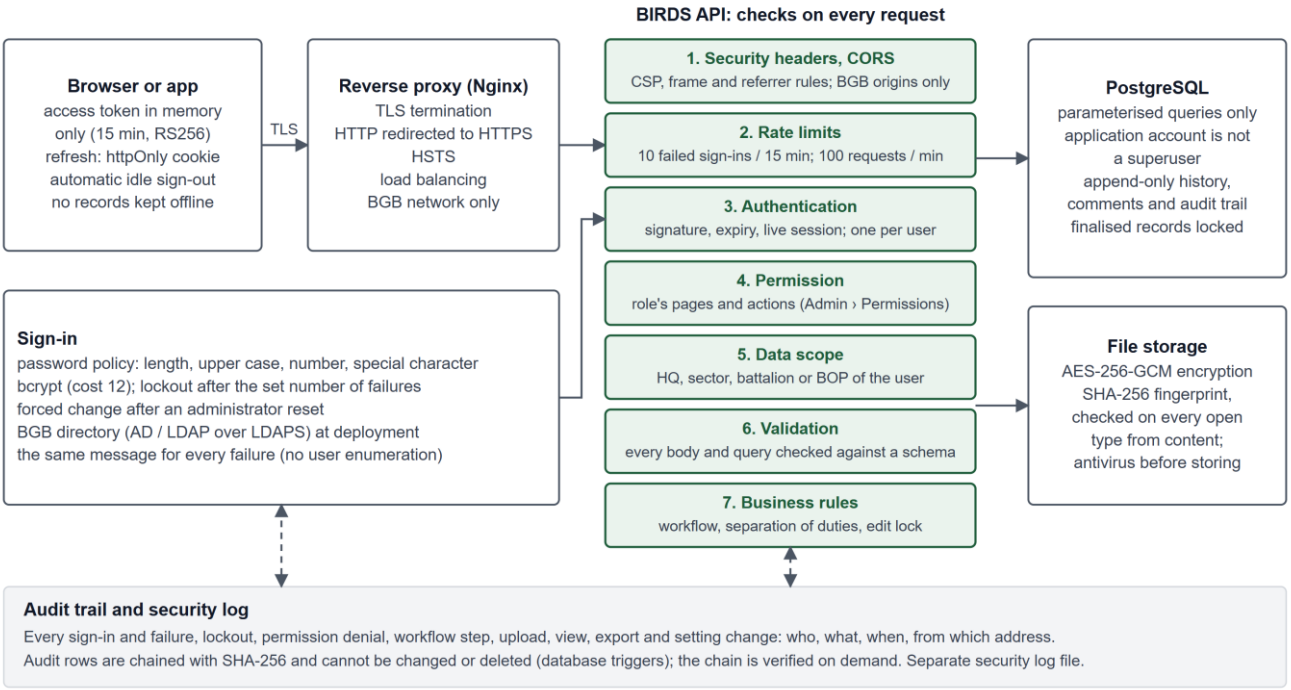


Figure 7. Security layers: every request passes each check in order; every outcome is audited (items 4, 24–29, 49)

Before production deployment an application security test and vulnerability assessment is carried out and the report provided; critical and high-risk findings are fixed before acceptance (item 50).

7. Infrastructure requirements

The infrastructure is dimensioned for the 500 concurrent users of item 38 with reserve capacity for growth, peak days and future modules, so that BGB does not need new hardware when use grows. Every figure below follows from a measurement or a stated assumption (7.1), and the purpose of each component, and what happens without it, is given in 7.3. A minimum configuration that meets item 38 without that reserve is shown alongside (7.2) so that BGB can compare costs. Final quantities are confirmed by the load test and agreed with BGB before procurement; virtual machines on BGB's existing virtualisation platform are suitable and normally the most economical option.

7.1 Workload basis

Basis	Value	Source
Concurrent active users	500	Item 38 (the highest figure in the specification)
User activity	One action every 20 seconds on average; about 4 server requests per action (the dashboard makes 8, a list 2, an incident page 3–4); 2 background checks per minute (notifications, new messages)	Usage model of the application

Basis	Value	Source
Average load	About 115 requests per second (500 × (4 ÷ 20 + 2 ÷ 60))	Calculation
Peak load	About 350 requests per second: three times the average, for shift changes and major incidents when most users open the dashboard at once	Assumption
Measured capacity	One BIRDS application process answered more than 650 requests per second of this request mix at every load level tested, 95 % of them within 100 ms, without errors, with up to 100 simultaneous clients (demonstration data, 15-10-2026, one process on a current laptop processor)	Measurement by the supplier
Planning capacity	350 requests per second per process: about half the measured value, allowing for slower server processor cores and for the larger data volume after years of use	Engineering margin
Incident volume	20,000 incidents per year (about 55 per day across all BOPs)	Assumption, to be confirmed by BGB
Evidence per incident	About 20 MB: 3 photographs (3 MB each), a video on one incident in three (30 MB), 1 document (0.5 MB)	Typical phone and office files
Database growth	About 10 GB per year: audit trail 4.7 GB (60 entries per user per day, 430 bytes each), messages 3.3 GB, notifications 1.9 GB, incident records, history and evidence records under 0.3 GB	Measured size per record × usage
File storage growth	About 0.5 TB per year: evidence 0.4 TB (20,000 × 20 MB) and voice messages 0.1 TB	Calculation
Retention	10 years (the default retention setting, configurable)	System setting (item 44)

7.2 Configuration

Component	Qty	Recommended: 500 users with reserve	Minimum: 500 users, no reserve
Reverse proxy / load balancer	2	2 vCPU, 4 GB RAM, 50 GB	2 vCPU, 4 GB RAM, 50 GB
Application server	2	8 vCPU, 16 GB RAM, 100 GB	4 vCPU, 8 GB RAM, 100 GB
Database server (primary + standby)	2	8 vCPU, 32 GB RAM, 500 GB SSD	4 vCPU, 16 GB RAM, 300 GB SSD

Component	Qty	Recommended: 500 users with reserve	Minimum: 500 users, no reserve
File storage (NAS or object storage)	1	2 TB usable, RAID, expandable	1.5 TB usable, RAID, expandable
Backup storage	1	4 TB usable	3 TB usable
Supporting services (antivirus, call relay, e-mail relay)	1	4 vCPU, 8 GB RAM, 100 GB	2 vCPU, 6 GB RAM, 100 GB
DR site	—	1 application server and 1 database server as above; file storage and backup storage equal to the main site	As the minimum above
Total, main site		40 vCPU, 112 GB RAM	22 vCPU, 62 GB RAM
Total, DR site		16 vCPU, 48 GB RAM	8 vCPU, 24 GB RAM

The minimum configuration meets item 38 with the same redundancy but without reserve: one application server then carries the full peak with little room for simultaneous large uploads and reports, and any growth in users needs additional servers. The recommended configuration is the one this proposal is based on.

7.3 Why each component is needed

Component	Why it is needed and how it is sized	Without it
Reverse proxy / load balancer (2)	The single entry point: terminates HTTPS (item 27), adds security headers, spreads requests over the application servers and keeps the 500 live connections used for messages and calls. TLS for about 350 requests per second uses a fraction of 2 vCPU; smaller virtual machines are not practical. Two, sharing one virtual address, because a single entry point would be a single point of failure (item 36).	Any fault or patch on a single proxy stops BIRDS for every user.

Component	Why it is needed and how it is sized	Without it
Application servers (2)	Run all of BIRDS: rules, permissions, workflow, audit, file handling. Sized so that one server alone carries the full peak with ample reserve: 8 vCPU run 8 processes, 2,800 requests per second of planning capacity against a peak of about 350. The rest covers operations outside the request mix: report and Excel generation, printing with evidence, encryption and decryption of evidence, the nightly archiving job. 16 GB: about 2.5 GB for 8 processes, plus uploads, which are checked in memory before storage (up to 10 files of 50 MB per request, so 10 simultaneous uploads can take 5 GB), plus the operating system.	With one server, a fault or an update stops BIRDS (items 36, 40). With smaller servers, simultaneous uploads and large reports slow everyone down.
Database servers (primary and standby)	Hold all records, history, audit trail and settings (item 30). 32 GB so that the active data (about 10 GB a year) stays in memory and searches and reports read from memory, keeping normal transactions under 3 seconds (item 39) as data grows. 8 vCPU for report aggregations and the connections of 16 application processes. 500 GB SSD: about 100 GB of data after 10 years, with room for growth, plus index rebuilds, transaction log and maintenance space; SSD because database response time depends on disk latency. The standby receives every change continuously and takes over if the primary fails (item 36; recovery point 15 minutes).	Without the standby, a database fault means restoring from backup: hours of downtime and possible loss of the day's records (items 36, 40).
File storage	Evidence files, voice messages and photos, encrypted by the application (items 10, 33). 2 TB covers about 4 years at 0.5 TB a year; expandable, so BGB buys capacity as it is used instead of for 10 years at once. RAID protects against a disk failure.	Evidence cannot be stored centrally; a disk failure loses evidence.
Backup storage	Automated backups (item 34) on storage separate from the live data, so that one failure cannot destroy both. Twice the live data: one full copy of the evidence (files never change, so each is copied once) and 30 days of daily database backups with the continuous transaction log for point-in-time recovery.	No recovery from deletion, corruption, ransomware or a storage failure.

Component	Why it is needed and how it is sized	Without it
Supporting services (1)	Antivirus engine that checks every upload (item 49; about 3 GB RAM for its signature database); relay for voice calls across network segments (Annex B, section 3); e-mail relay if BGB has none. Light services, so one virtual machine.	Not duplicated: a fault affects upload scanning, calls or e-mail for its duration; incident recording, review and reporting continue.
DR site	Restores service if the main site is lost (item 35). One application server is enough because one server carries the full peak; the database is the same size as the main one so that it can take over; storage holds the replica of the files and a copy of the backups.	Loss of the main site (fire, power, flood) stops BIRDS until it is rebuilt.

Network: about 20 Mbit/s for 500 users' normal use, so at least 100 Mbit/s to the data centre to keep evidence uploads and downloads quick (a 30 MB video in under 3 seconds); 1 Gbit/s between the servers; at least 20 Mbit/s between the main and DR sites for continuous replication. HTTPS (443) from users to the proxy; internal ports between proxy, application, database and storage only.

Operating system: Ubuntu Server LTS (22.04 or 24.04) or another Linux distribution supported by BGB. Client devices: current Google Chrome, Microsoft Edge or Mozilla Firefox. Infrastructure for the value additions (voice calls, push notifications, mobile use) is listed in Annex B, section 3.

Map display (items 8 and 18): BIRDS needs no internet and no separate map server for its maps. The application servers hold one file with the vector map of Bangladesh and the border strip (OpenStreetMap data, about 340 MB, street level), and the browser draws the map from it, with place names in English or Bangla. The demonstration runs the same way, without internet. Newer map data is installed by replacing the file (one command, on any computer with internet).

8. Availability, backup and disaster recovery

8.1 High availability

Two or more application servers behind the load balancer and a standby database remove single points of failure for the critical components (item 36). Because the application keeps no state in memory, a failed application server is simply taken out of rotation. The database standby is promoted if the primary fails. Planned updates are applied one server at a time. Together with monitoring and maintenance procedures this supports the 99.5 % availability target of item 40.

8.2 Backup

a. Automated, scheduled backup of the database and of the evidence files (item 34); frequency configurable in the Admin module (the setting already exists).

b. Continuous archiving of the database transaction log (WAL) between full backups, so the database can be restored to any point in time.

c. Backups encrypted and copied to separate backup storage and to the DR site; each run recorded in the audit trail.

d. Restore tests at regular intervals to prove that backups are usable.

8.3 Disaster recovery

The database is replicated continuously to the DR site and evidence files are synchronised to DR storage (item 35). If the main site is lost, the DR database is promoted, the application is started at the DR site, and users are redirected. The step-by-step procedure is delivered in the Backup/Recovery Procedure and rehearsed with BGB.

Recovery target	Value
Recovery point objective (data loss at most)	15 minutes
Recovery time objective (service restored within)	4 hours

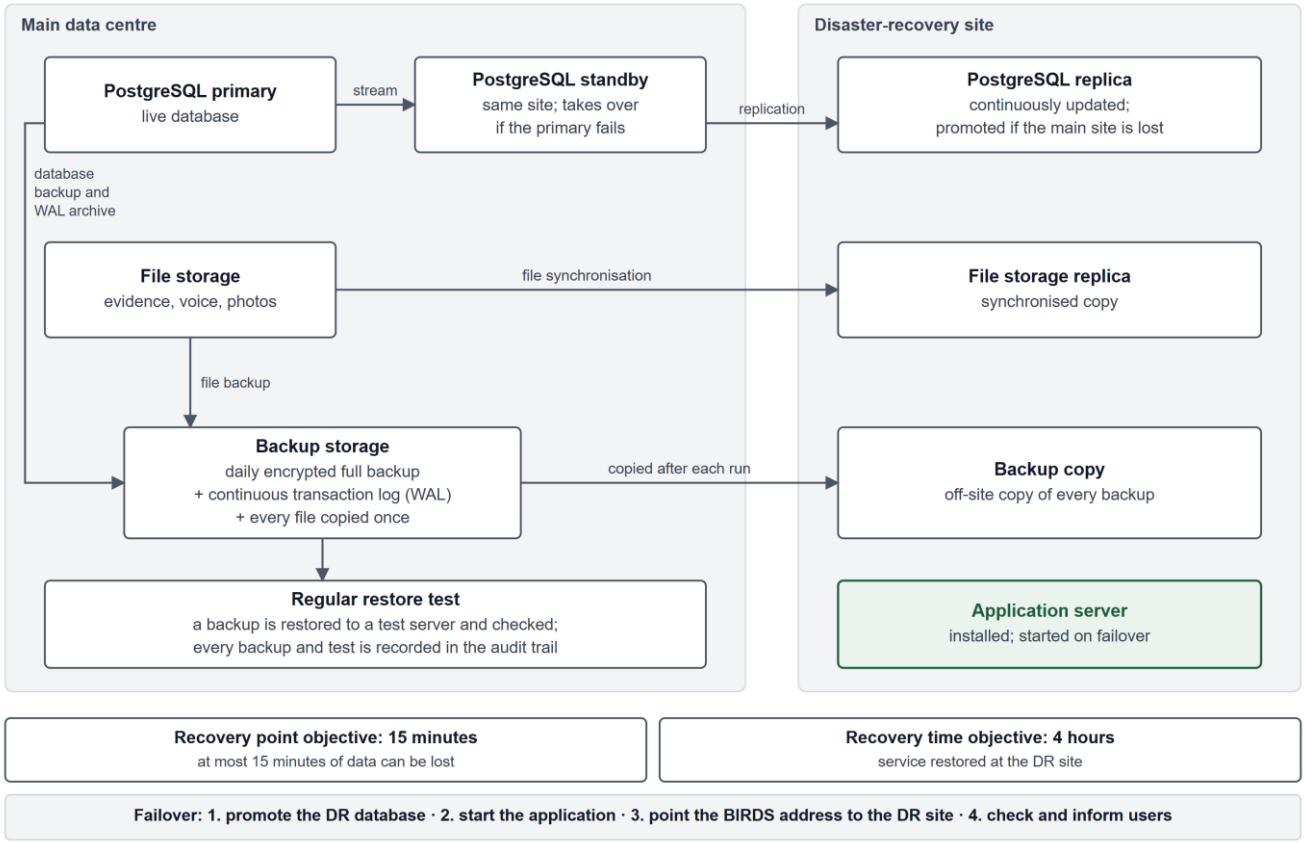


Figure 8. Backup, replication and disaster recovery, set up during implementation (items 34, 35, 36)

9. Performance and scalability

Application servers can be added behind the load balancer, the database has indexes for every common search, every list is paged, and files are kept outside the database, so users, incidents, attachments, units and history can grow without redesign (item 37). Before acceptance a load test is run at 100, 250 and 500 concurrent users with realistic incident, search and report traffic, on the installed BGB infrastructure. Acceptance criterion: normal

transactions respond within 3 seconds at 500 concurrent users, excluding large file transfers and complex reports (items 38, 39). The load-test report is provided.

10. Integration and future expansion

a. REST API (/api/v1) secured with the same tokens and permission checks as the user interface, and documented in OpenAPI 3.1 with an interactive console (item 31).

b. Directory sign-in: users sign in with their BGB directory account over LDAPS; directory groups are mapped to BIRDS roles and scopes; accounts are created on first sign-in; local accounts can be kept as a fallback if BGB wishes (item 32).

c. Future systems (item 58): mobile applications, surveillance systems, digital evidence systems and other BGB systems connect through the documented API; the system already includes realtime messaging, voice calls and push notifications and is installable on phones, showing this extensibility.

11. Implementation methodology and work plan

Implementation follows the phases below. Durations are indicative and will be fixed in the work plan agreed at project start.

#	Phase	Main activities	Weeks
1	Mobilisation and requirement confirmation	Kick-off, confirm workflow, roles, hierarchy, categories, report formats and notification rules with BGB	1–2
2	Configuration and customisation	Configure hierarchy, categories, roles and policies; customise forms, reports and workflow to BGB requirements	2–5
3	Infrastructure and installation	Install servers, database replication, storage, proxy and TLS at the main and DR sites	4–6
4	Integration	Directory (AD/LDAP), SMTP relay, antivirus engine, current offline map data	5–6
5	Data migration (if applicable)	Import existing incident records and users after cleansing and validation	5–7
6	Testing	System testing, security test and vulnerability assessment, load test, backup/restore and DR rehearsal	6–8
7	User acceptance test	Run the approved UAT test cases with BGB; fix findings	8–9
8	Training	Administrator, operator and end-user training	9–10
9	Go-live and commissioning	Production deployment, hypercare support, handover of source code and documents	10



Figure 9. Indicative work plan (item 56)

12. Testing and quality assurance

a. Automated tests: the server has an automated test suite (171 tests at the time of this proposal) covering authentication, permissions, workflow, evidence, audit, reports and messaging, plus a check that the API documentation matches the code; the main user journeys are tested in Chrome, Edge and Firefox and on phone, tablet and desktop screen sizes.

b. Security: secure code review, dependency vulnerability audit, OWASP Top 10 assessment and vulnerability scan before production (items 49, 50).

c. Performance: load test at 100, 250 and 500 concurrent users (items 38, 39).

d. Acceptance: functional acceptance test (UAT) against the approved requirements and test cases; final acceptance on its successful completion (item 57).

13. Training

Hands-on training is given on the installed system with training material in English and Bangla (item 53).

Audience	Duration	Content
System administrators	2 days	Installation overview, users and roles, permissions, hierarchy and categories, system settings, backup and restore, logs and monitoring, troubleshooting
Designated operators (BOP / Battalion)	1 day	Incident entry, evidence, GPS location, drafts and submission, responding to returns, search and tracking

Audience	Duration	Content
Supervisory officers (Sector / Verifier / HQ)	1 day	Review, return, verify, approve, reject, dashboard, GIS map, analytics, reports and export
Auditors	½ day	Audit trail search, chain verification, exports

14. Documentation

The following documents are delivered in English (item 52), in addition to the source code, database schema and configuration (item 51):

Document	Contents
System Administration Manual	Users and roles, permissions, hierarchy, categories, all configuration settings, logs, monitoring, account lifecycle
User Manual	Sign-in, dashboard, filing incidents, evidence, workflow actions, search, GIS, reports and export, notifications, password change
Installation / Deployment Guide	Prerequisites, server setup, database and replication, storage, proxy and TLS, environment settings, updates
API Documentation	OpenAPI 3.1 description of every endpoint (already available in the demo at /api/docs)
Database Documentation	Entity-relationship diagram, table and column dictionary, integrity rules
Backup / Recovery Procedure	Backup schedule, restore, point-in-time recovery, DR failover, recovery targets, test checklist

15. Warranty and support

One year of warranty and technical support from acceptance (item 54), covering bug fixing, security patches, configuration support and troubleshooting. Proposed response times:

Priority	Example	Response / resolution target
Critical	System unavailable or data at risk	2 hours / 8 hours
High	A main function unusable for many users	4 hours / 2 working days
Medium	A function impaired; workaround exists	1 working day / 5 working days
Low	Question, cosmetic issue	2 working days / next release

Support is provided by telephone, e-mail and on-site visits, from Sunday to Thursday, 9:00 to 17:00 Bangladesh time; critical incidents are accepted at any time.

Address : House 5, Road 2, Block C, Banasree, Rampura, Dhaka 1219
Phone : 01755253505, 01730357123, 01701072702
E-mail : graceinternational98@gmail.com

16. BGB responsibilities and assumptions

a. BGB provides the servers (physical or virtual), storage, network, TLS certificate and DR site described in section 7, and secure access for the supplier's engineers during implementation and support.

b. BGB provides directory (AD/LDAP) connection details and a service account, an SMTP relay, and names the antivirus product to be integrated.

c. BGB nominates officers to confirm requirements, take part in UAT and attend training.

d. Data migration covers existing records supplied by BGB in an agreed format.

e. Infrastructure for the value additions is listed in Annex B, section 3.

17. Value additions

In addition to the specification, the offered system includes secure messaging, voice messages and calls, push notifications, a progressive web app installable on phones and tablets from the browser, evidence integrity checks, a hash-chained audit trail, formal printouts with reference numbers, a work queue, duplicate detection and a full Bangla interface, among others. They are listed in Annex B, are included in the quoted price and do not form a condition of this offer.

Name of authorised signatory:	
Designation:	
Signature and seal:	
Date:	18/10/2026